Variational Year-Style Translation: A Single-Generator Conditional CycleGAN for Re-Dating Object Imagery

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Abstract

We study *visual re-dating*: rendering an object crop as it would have appeared in a chosen target year while preserving its content. A faithful multi-domain CycleGAN would require a separate generator/discriminator pair for every ordered pair of year buckets, which scales quadratically with the number of buckets. We instead condition a *single* generator on a target year through a *variational year code*: each year is a small Gaussian in a latent space, sampled and mapped to a style vector that modulates the generator via adaptive instance normalization. Year identity is enforced by an auxiliary year classifier in the discriminator, while a Kullback–Leibler term keeps the latent compact and continuous, enabling deterministic targeting, intra-year style variation, and interpolation to unseen in-between years. The model is trained with cycle-consistency, identity, adversarial, classification, and KL objectives. This section details the formulation; experiments are reported in the following section.

1 Method

Our goal is to translate the apparent *date* of an object photograph: given a cropped object and a target year, re-render the object so that its low-level appearance (film response, colour, grain, lighting, and period-specific styling) matches that target year, while preserving object identity and pose. We treat year as a categorical domain and adopt the structure of CycleGAN [Zhu et al., 2017]—two mapping directions tied by cycle consistency and two adversarial signals—but avoid instantiating one network per ordered pair of domains. Instead, following the multi-domain philosophy of StarGAN [Choi et al., 2018], we use a single generator conditioned on a target year, and we make the conditioning *variational* so that years live in a compact, continuous latent space. Figure 1 summarises the training step.

1.1 Problem formulation and notation

Let $\mathcal{X} \subset \mathbb{R}^{3\times H\times W}$ be the space of object crops, normalised to $[-1, 1]$. Each crop x carries an *origin* year label $y_o \in \{1, \dots, K\}$, where the K labels index contiguous five-year buckets spanning 1930–1999 ($K = 14$). We seek a single conditional generator

$$G:\ \mathcal{X} \times \mathbb{R}^{d_w} \ \to\ \mathcal{X}, \qquad \hat{x} \ =\ G(x, w), \tag{1}$$

that re-renders x in a target year encoded by a *style vector* $w \in \mathbb{R}^{d_w}$. Writing w_y for a style vector that represents year y, the translation of x to year y is $G(x, w_y)$; we abbreviate this by $G(x, y)$ when the style network is clear from context. A vanilla two-domain CycleGAN models a single pair $X \leftrightarrow Y$ with two generators and two discriminators; the faithful K-domain generalisation would need $\mathcal{O}(K^2)$ generators. By conditioning one generator on w_y we reduce this to a single network shared across all years.

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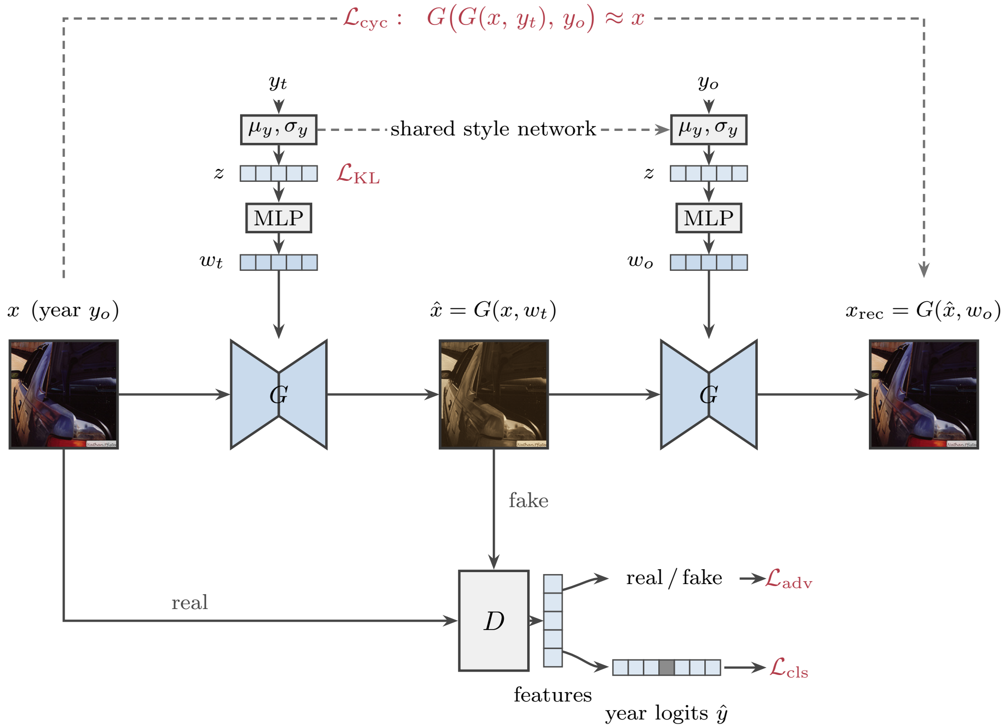

Figure 1: One training step of the year-conditional CycleGAN. A shared style network draws a year code $z \sim \mathcal{N}(\mu_y, \sigma_y^2)$ for the target and origin years and maps it to style vectors w_t, w_o that modulate the shared generator G via adaptive instance normalization. The forward cycle $x \to \hat{x} = G(x, w_t) \to x_{\text{rec}} = G(\hat{x}, w_o)$ is constrained by cycle consistency, $G(G(x, y_t), y_o) \approx x$. The discriminator D produces a feature vector from which two heads read a real/fake score (adversarial loss $\mathcal{L}_{\text{adv}}$) and a year posterior (classification loss $\mathcal{L}_{\text{cls}}$); the year code is regularised by $\mathcal{L}_{\text{KL}}$.

1.2 Year as a variational style

We represent each year not as a point but as a compact distribution in a latent space, the probabilistic analogue of a per-year proxy. Concretely, a style network S holds two embedding tables, $\mu \in \mathbb{R}^{K \times d_z}$ and $\log \sigma^2 \in \mathbb{R}^{K \times d_z}$, defining a per-year Gaussian

$$q(z \mid y) \;=\; \mathcal{N}\big(\mu_y,\; \mathrm{diag}(\sigma_y^2)\big). \tag{2}$$

A code is drawn with the reparameterisation trick [Kingma and Welling, 2014],

$$z \;=\; \mu_y + \sigma_y \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{3}$$

and mapped to a style vector by a small multilayer perceptron, $w_y = \mathrm{MLP}(z) \in \mathbb{R}^{d_w}$. Setting $\epsilon = \mathbf{0}$ (i.e. $z = \mu_y$) gives a *deterministic* style for year y, while sampling $\epsilon \neq \mathbf{0}$ yields intra-year style variation.

Conditioning the generator. The generator is a residual encoder–decoder [Johnson et al., 2016]: a downsampling stem, a stack of residual blocks, and an upsampling head with reflection padding throughout. The style vector w enters only through the residual blocks, via adaptive instance normalisation [Huang and Belongie, 2017]. For a feature map h with per-channel mean $\mu(h)$ and standard deviation $\sigma(h)$,

$$\mathrm{AdaIN}(h, w) \;=\; \big(\mathbf{1} + \gamma(w)\big) \odot \frac{h - \mu(h)}{\sigma(h)} \;+\; \beta(w), \tag{4}$$

where $(\gamma(w), \beta(w))$ are produced by an affine layer per block and are initialised near zero so that each block starts close to the identity. Because w modulates only normalisation statistics, content is carried by the convolutional features while *year* is injected as a global style—exactly the disentanglement we require for re-dating.

Latent regularisation. To keep each year's latent compact and the overall space continuous (so that years can be interpolated), we regularise $q(z \mid y)$ towards a standard normal prior with a Kullback–Leibler penalty,

$$\mathcal{L}_{\mathrm{KL}}(y) \;=\; \mathrm{KL}\big(q(z \mid y) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\big) \;=\; \tfrac{1}{2} \sum_{i=1}^{d_z} \big(\mu_{y,i}^2 + \sigma_{y,i}^2 - \log \sigma_{y,i}^2 - 1\big). \tag{5}$$

We use a deliberately small weight on this term: *separation* between years is not provided by the KL (which would, on its own, collapse the years onto the prior) but by the year classifier described next, which forces z to carry year identity. The KL only enforces compactness and continuity.

Initialising the year geometry from learned proxies. The latent means $\{\mu_y\}$ need not start from noise. A representation-learning model trained on the same crops with a metric (proxy) loss over years produces a set of year *proxies*—centroids $c_y \in \mathbb{R}^d$ whose geometry already encodes the order and relative spacing of decades. We optionally initialise $\mu_y \leftarrow \bar{c}_y$, where $\bar{c}_y$ is the ℓ_2-normalised proxy (the proxy loss is cosine-based, so only direction is meaningful), matching d_z to the proxy dimension; σ_y is initialised small as before. The generator then starts from the *true* year manifold rather than a random one. Section 2 evaluates whether this geometric prior improves convergence, latent ordering, and extrapolation to held-out years.

1.3 Discriminator with an auxiliary year classifier

The discriminator D is a fully convolutional network that maps an image to a spatial feature tensor, from which two linear heads branch (Figure 1, bottom). The first head outputs an unnormalised real/fake field used for the adversarial signal; the second is a *year classifier* that, after spatial pooling, produces a posterior over the K year buckets, $D_{\mathrm{cls}}(y \mid x)$. The classifier plays the role of the StarGAN domain classifier [Choi et al., 2018]: it is what actually *instantiates* a target year, since the generator is rewarded only when its output is classified as the requested year. We use a least-squares adversarial objective [Mao et al., 2017] for stable training,

$$\mathcal{L}_{\mathrm{adv}}^{D} = \tfrac{1}{2}\mathbb{E}_x\big[(D_{\mathrm{src}}(x) - 1)^2\big] + \tfrac{1}{2}\mathbb{E}_{\hat{x}}\big[D_{\mathrm{src}}(\hat{x})^2\big], \qquad \mathcal{L}_{\mathrm{adv}}^{G} = \tfrac{1}{2}\mathbb{E}_{\hat{x}}\big[(D_{\mathrm{src}}(\hat{x}) - 1)^2\big], \tag{6}$$

where D_{src} denotes the real/fake head. To stabilise the discriminator we draw $\hat{x}$ for the D update from a buffer of previously generated images [Shrivastava et al., 2017].

1.4 Training objective

Given a real crop x with origin year y_o and a target year y_t sampled uniformly at random, we obtain style vectors $w_t = \mathrm{MLP}(z_t)$ and $w_o = \mathrm{MLP}(z_o)$ from Eq. (3), the forward translation $\hat{x} = G(x, w_t)$, the cyclic reconstruction $x_{\mathrm{rec}} = G(\hat{x}, w_o)$, and the identity mapping $G(x, w_o)$.

Year classification. The discriminator's classifier is trained on *real* images with their true origin year, while the generator is trained so that its output is classified as the *target* year:

$$\mathcal{L}_{\mathrm{cls}}^{D} = \mathbb{E}_{x, y_o}\big[-\log D_{\mathrm{cls}}(y_o \mid x)\big], \qquad \mathcal{L}_{\mathrm{cls}}^{G} = \mathbb{E}_{x, y_t}\big[-\log D_{\mathrm{cls}}(y_t \mid \hat{x})\big]. \tag{7}$$

Cycle consistency and identity. Translating to the target year and back to the origin year should recover the input, and translating to the origin year should be a no-op:

$$\mathcal{L}_{\mathrm{cyc}} = \mathbb{E}\big[\, \|G(G(x, y_t), y_o) - x\|_1 \,\big], \qquad \mathcal{L}_{\mathrm{idt}} = \mathbb{E}\big[\, \|G(x, y_o) - x\|_1 \,\big]. \tag{8}$$

Both use an ℓ_1 reconstruction penalty. The cycle term is the mechanism that ties content across years; the identity term discourages spurious changes when the requested year already matches the input.

Full objective. The discriminator and the generator (jointly with the style network) are optimised by alternating descent on

$$\mathcal{L}_D = \mathcal{L}^D_{\mathrm{adv}} + \lambda_{\mathrm{cls}}\, \mathcal{L}^D_{\mathrm{cls}}, \tag{9}$$

$$\mathcal{L}_G = \mathcal{L}^G_{\mathrm{adv}} + \lambda_{\mathrm{cls}}\, \mathcal{L}^G_{\mathrm{cls}} + \lambda_{\mathrm{cyc}}\, \mathcal{L}_{\mathrm{cyc}} + \lambda_{\mathrm{idt}}\, \mathcal{L}_{\mathrm{idt}} + \lambda_{\mathrm{kl}} \left(\mathcal{L}_{\mathrm{KL}}(y_t) + \mathcal{L}_{\mathrm{KL}}(y_o)\right). \tag{10}$$

The style network receives gradients only through $\mathcal{L}_G$: from the adversarial and classification terms (which shape w so that the requested year is realised) and from the KL term (which regularises the latent). Crucially, the classifier gradient is what keeps years distinct, so a small λ_{kl} does not collapse the latent.

1.5 Optimisation and architecture

Both networks use instance normalisation [Ulyanov et al., 2016] and weights initialised from $\mathcal{N}(0, 0.02^2)$. The generator has 64 base channels and 9 residual blocks; the style network uses $d_z = d_w = 64$. We optimise with Adam [Kingma and Ba, 2015] (β_1=0.5, β_2=0.999) under a two time-scale update rule [Heusel et al., 2017], giving the discriminator a faster learning rate than the generator. The learning rate is held constant for the first half of training and then decayed linearly to zero [Zhu et al., 2017]. For sampling we maintain an exponential moving average of the generator and style-network weights. Unless otherwise stated we use λ_{cyc}=10, λ_{idt}=5, λ_{cls}=1, and λ_{kl}=10^{-2}.

1.6 Inference: targeting, sampling, and interpolation

At test time the variational code supports three modes. *Deterministic targeting* of a year y uses the latent mean, $w_y = \mathrm{MLP}(\mu_y)$, which gives a single canonical rendering. *Stochastic* translation samples $z \sim q(z \mid y)$ to produce diverse in-year styles for a fixed content. Finally, because the KL term keeps the latent continuous, *unseen or in-between years* are reachable by interpolating the latent means,

$$w_{a\to b}(t) = \mathrm{MLP}\big((1-t)\,\mu_a + t\,\mu_b\big), \qquad t \in [0,1], \tag{11}$$

yielding a smooth appearance trajectory between two year buckets without retraining. A single forward pass through G then re-renders any input crop at the chosen point on the year axis.

2 Experimental setup

Visual re-dating has *no paired ground truth*: a given object was never photographed in two different years, so a translation cannot be compared to a pixel-aligned reference. Every quantity below is therefore either a per-image descriptor whose *distribution* we compare to real photos of the target year, or a perceptual difference between matched summaries. Because the year signal in this corpus is, to first order, *photographic colour* (palette, white balance, contrast, sepia-vs-cool channel correlation, film grain), we lead with colour-fidelity metrics from the colour-transfer and colorization literature, and complement them with an independent date oracle, content preservation, and a downstream train-on-synthetic test. Empty cells (xx) in the tables are reported in the camera-ready.

2.1 Data and protocol

We use object crops from *Date Estimation in the Wild* (DEW) [Müller et al., 2017], dated photographs spanning 1930–1999. Crops are taken from detector boxes (Section 1.1) for several object categories (CAR, HOUSE, and person classes), resized to 128^2 and normalised to $[-1, 1]$; years are quantised into the $K = 14$ five-year buckets of Section 1.1. We use the official held-out evaluation index for the test split, with no image shared between train and test. To probe extrapolation we additionally hold out one interior bucket (e.g. 1965–1969) from training and reach it only by latent interpolation (Section 1.6).

Independent date oracle. Re-dating accuracy is judged by a date classifier D^* trained from scratch on real crops with cross-entropy over the K buckets, and *never* used to supervise the GAN: it is distinct from the discriminator's year head and from the proxy checkpoint that seeds the latent means, so it does not measure the signal the generator was trained to fool. We first report D^*'s own accuracy on the real test split: because a translation can be recognised as year y_t no more reliably than a real photo of year y_t, the oracle's accuracy is the *ceiling* for any model's re-dating accuracy, and all generated numbers are read relative to it.

2.2 Baselines

We compare against **(i)** *colour-transfer heuristics*—Reinhard colour transfer [Reinhard et al., 2001] and per-channel histogram matching to the target bucket's statistics, plus greyscale and fixed sepia—which only touch global colour and lower-bound what appearance statistics alone can achieve; **(ii)** *StarGAN* [Choi et al., 2018], one generator conditioned on a one-hot year with no variational latent, isolating the contribution of the year code; **(iii)** a *free-latent* variant after StarGAN-v2 [Choi et al., 2020], drawing the style from a single global Gaussian rather than a per-year $q(z \mid y)$; and **(iv)** a non-scalable *per-pair CycleGAN* [Zhu et al., 2017] on a single bucket pair, a strong reference for those two directions. The colour-transfer baselines are essential because, as Section 2.3 makes explicit, a method can match a bucket's *mean* colour statistics without producing a convincingly period-correct image; the baselines reveal whether the GAN does more than global colour matching.

2.3 Metrics

(a) Colour fidelity (primary). Treating re-dating as a colour/tone transfer, we report, per target bucket and averaged over buckets: a *colour-FID*, the Fréchet distance between Gaussians fit to an 11-dimensional photographic-colour signature (CIELAB mean/std, chroma, RG/RB/GB channel correlation, and high-frequency grain energy) of the generated vs. real images—the colour-space analogue of FID, but interpretable per attribute; the *palette* ΔE_{00}, a CIEDE2000 [Sharma et al., 2005] perceptual difference between the tonal palettes (lightness-binned mean Lab colours) of the two sets; the *hue EMD*, an Earth Mover's Distance [Rubner et al., 2000] between hue histograms; and the *colourfulness* [Hasler and Süsstrunk, 2003] trend, the correlation between the generated and real colourfulness-versus-decade curves (real photographs grow more colourful from 1930 to 1999; a faithful model reproduces this monotonic trend). We stress that mean-palette metrics (ΔE_{00}, colourfulness trend) can be *minimised by construction* by a colour-transfer baseline fitted to bucket statistics; the distributional colour-FID and the oracle below are what separate genuine re-dating from palette matching. **(b) Re-dating accuracy.** With $\hat{y} = \arg\max_k D^*(k \mid G(x, y_t))$ we report top-1 target accuracy $\mathrm{acc@0}$, the off-by-one tolerance $\mathrm{acc@1} = \Pr(|\hat{y} - y_t| \leq 1)$, the mean absolute date error MAE in years, and the source→target confusion matrix (the diagonal should be strong in *all* directions). **(c) Content preservation.** Cycle fidelity via luminance-SSIM [Wang et al., 2004] between x and x_{rec} (structure must survive a colour change while the chroma is free to move), the identity error $\|G(x, y_o) - x\|_1$, and the cosine similarity of the year-agnostic object embeddings $\phi(x)$ and $\phi(G(x, y_t))$ from the representation encoder. **(d) Realism.** FID [Heusel et al., 2017]/KID [Bińkowski et al., 2018] between translations into bucket k and real bucket-k images; when an ImageNet feature extractor is unavailable the distributional colour-FID of (a) serves as a domain-specific substitute. **(e) Variational year code.** Interpolation *monotonicity* (Spearman $\rho(t, \hat{y})$ along $w_{a\to b}(t)$), latent *ordering* (Spearman correlation between the leading principal coordinate of $\{\mu_y\}$ and the true year), intra-year *diversity* (mean pairwise LPIPS [Zhang et al., 2018] among stochastic samples), and *extrapolation* (re-dating accuracy on the held-out interior bucket reached only by interpolation).

2.4 Downstream temporal consistency (TSTR)

As a downstream check of whether the year-swap encodes *transferable* temporal appearance, we train a date estimator only on synthetic year-swapped crops—each real crop translated to a uniformly

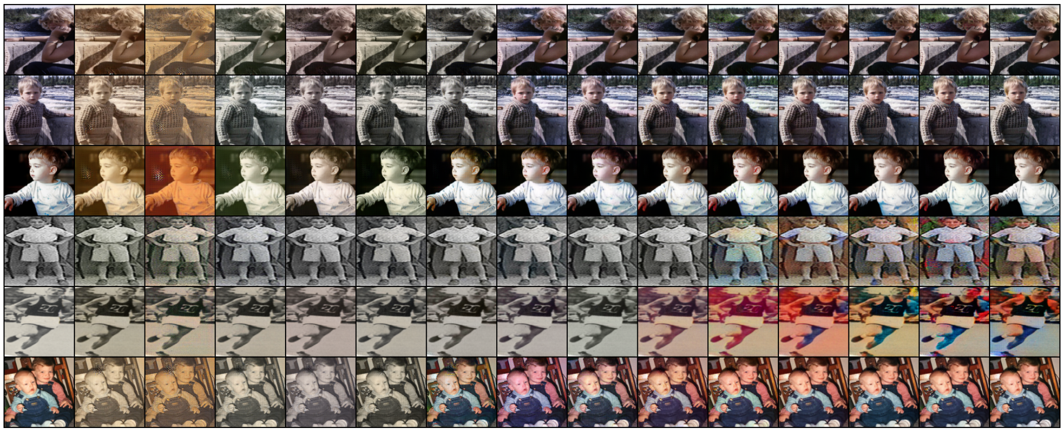

Figure 2: Qualitative re-dating (preliminary model). Each row is a source crop (leftmost) followed by its translation $G(x, y_t)$ to each of the $K = 14$ five-year buckets, from 1930–34 (left) to 1995–99 (right), using the deterministic mean style μ_{y_t}. Tone and palette shift from warm, low-contrast early-century renderings to cooler, more saturated modern ones while object content is preserved.

Table 1: Main comparison on the DEW re-dating benchmark. Colour fidelity is the primary axis; re-dating accuracy is bounded by the oracle ceiling (top row). Arrows give the preferred direction; † marks the non-scalable per-pair reference (trained bucket pair only). Numbers filled in the camera-ready.

	Colour fidelity			Date change		Content
Method	cFID↓	ΔE_{00}↓	hue↓	acc@1↑	MAE↓	SSIM↑
oracle ceiling / chance	–	–	–	xx / xx	xx	–
Sepia / greyscale	xx	xx	xx	xx	xx	xx
Reinhard transfer [Reinhard et al., 2001]	xx	xx	xx	xx	xx	xx
Histogram matching	xx	xx	xx	xx	xx	xx
Per-pair CycleGAN† [Zhu et al., 2017]	xx	xx	xx	xx	xx	xx
StarGAN [Choi et al., 2018]	xx	xx	xx	xx	xx	xx
Free-latent [Choi et al., 2020]	xx	xx	xx	xx	xx	xx
Ours (variational year code)	xx	xx	xx	xx	xx	xx

random target year y_t and labelled with y_t—and test it on *real* images (train-on-synthetic, test-on-real, TSTR [Shmelkov et al., 2018]). A small gap to a model trained on real data indicates that synthetic year cues match real ones. Two controls make this rigorous: an *identity control* that relabels untranslated crops with random target years (which must collapse to chance, proving the GAN rather than residual content carries the year signal), and an *augmentation* run (real+synthetic) measuring whether re-dated crops are useful training data.

2.5 Evaluating the proxy initialisation

Finally we test whether seeding μ_y with the learned proxy centroids (Section 1.2) helps, using matched runs—identical seed, architecture, and budget—that differ *only* in the initialisation of μ_y (random vs. ℓ_2-normalised centroids). We report re-dating accuracy and (colour-)FID versus training step (convergence), the latent-ordering ρ and interpolation monotonicity at convergence (geometry), extrapolation accuracy on the held-out bucket, and all of these at $10/25/50\%$ of the training data (efficiency), expecting the geometric prior to help most in the low-data and extrapolation regimes.

Table 2: Ablations. "ord. ρ" is latent year-ordering; "extrap." is acc@1 on the held-out bucket.

Variant	cFID↓	SSIM↑	ord. ρ↑	extrap.↑
Ours (centroid init)	xx	xx	xx	xx
random μ init	xx	xx	xx	xx
− KL	xx	xx	xx	xx
− identity	xx	xx	xx	xx
− colour head	xx	xx	xx	xx
− colour classifier	xx	xx	xx	xx

Table 3: TSTR: date estimator trained on each set, tested on real.

Training set	acc@0↑	acc@1↑
real (TRTR ref.)	xx	xx
synthetic (TSTR)	xx	xx
identity (control)	xx	xx
real + synthetic	xx	xx
chance	xx	xx

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